

JEE MAIN REVISION MASTER: INVERSE TRIGONOMETRIC FUNCTIONS 2026 2 QUESTIONS

3.33% WEIGHTAGE

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 22 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Inverse Trigonometric Functions — Standard Calculations & Problem Archetypes	15 PYQs	Quantitative Analysis
Moderate	Inverse Trigonometric Functions — Core Principles & Governing Laws	4 PYQs	Core Principle
Moderate	Inverse Trigonometric Functions — Applied Cases, Reactions & Traps	4 PYQs	Applied Topics

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Inverse Trigonometric Functions — Standard Calculations & Problem Archetypes	$\cos^{-1}(x) + \cos^{-1}(y) + \cos^{-1}(z) = 3\pi$, $\quad \cos(\sin^{-1}\frac{\pi}{3} + \cos^{-1}\frac{\pi}{3})$, $\quad \tan^{-1}(\frac{1}{1+k(k+1)}) = \tan^{-1}(k+1) - \tan^{-1}k$
Inverse Trigonometric Functions — Core Principles & Governing Laws	$\cos[\cot^{-1}(-\sqrt{3}) + \frac{\pi}{6}]$, $\quad \cos^{-1}x + \sin^{-1}y$, $\quad 2\sin^{-1}x + 3\cos^{-1}x$
Inverse Trigonometric Functions — Applied Cases, Reactions & Traps	$a + \frac{\pi}{2} < 2\tan^{-1}x + 3\cot^{-1}x < b$, $\quad \cos^{-1}[x]$, $\quad y = \cos^{-1}[x]$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Inverse Trigonometric Functions — Standard Calculations & Problem Archetypes (Very High Yield — 15 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Inverse Trigonometric Functions.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): " $\tan^{-1}\left(\frac{1}{1+1 \cdot 2}\right) + \tan^{-1}\left(\frac{1}{1+2 \cdot 3}\right) + \dots + \tan^{-1}\left(\frac{1}{1+n(n+1)}\right) = \dots$ " [Ans: A]
- [2025 Shift] Q#4 (MCQ): " $2\cos^{-1}x = \sin^{-1}(2x\sqrt{1-x^2})$ is valid for all values of 'x' satisfying..." [Ans: N/A]
- [2024 Shift] Q#1 (MCQ): "Let $f: R \rightarrow R$ be given $f(x) = \tan x$. Then, $f^{-1}(1)$ is..." [Ans: N/A]

3.2 Inverse Trigonometric Functions — Core Principles & Governing Laws (Moderate Yield — 4 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Inverse Trigonometric Functions.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Inverse Trigonometric Functions.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Inverse Trigonometric Functions PYQs.

Representative Source PYQs:

- [2025 Shift] Q#3 (MCQ): " $\sec^2(\tan^{-1}2) + \operatorname{cosec}^2(\cot^{-1}3) = \dots$ " [Ans: N/A]
- [2024 Shift] Q#3 (MCQ): "If $2\sin^{-1}x - 3\cos^{-1}x = 4, x \in [-1, 1]$, then $2\sin^{-1}x + 3\cos^{-1}x$ is equal to..." [Ans: N/A]
- [2021 Shift] Q#3 (MCQ): " $\cos[\cot^{-1}(\sqrt{3}) + (\pi/6)]$ is equal to..." [Ans: N/A]

3.3 Inverse Trigonometric Functions — Applied Cases, Reactions & Traps (Moderate Yield — 4 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Inverse Trigonometric Functions.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2026 Shift] Q#2 (MCQ): "If $\sin^{-1}x + \sin^{-1}y = (\pi/2)$, then x^2 is equal to..." [Ans: A]
- [2022 Shift] Q#2 (MCQ): "Domain $\cos^{-1}[x]$ is, where $[]$ denotes a greatest integer function..." [Ans: N/A]
- [2020 Shift] Q#2 (MCQ): "The domain of the function defined by $f(x) = \cos^{-1}\sqrt{x-1}$ is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

$$\tan^{-1}\left(\frac{1}{1+1 \cdot 2}\right) + \tan^{-1}\left(\frac{1}{1+2 \cdot 3}\right) + \dots + \tan^{-1}\left(\frac{1}{1+n(n+1)}\right) =$$

- (A) $\tan^{-1}\left(\frac{n}{n+2}\right)$ [Correct]
- (B) $\tan^{-1}\left(\frac{n+1}{n}\right)$
- (C) $\tan^{-1}\left(\frac{n}{n+1}\right)$
- (D) $\tan^{-1}\left(\frac{n+2}{n}\right)$

Correct Answer: A

Solution Analysis: Given sum is $\tan^{-1}\left(\frac{1}{1+1 \cdot 2}\right) + \tan^{-1}\left(\frac{1}{1+2 \cdot 3}\right) + \dots + \tan^{-1}\left(\frac{1}{1+n(n+1)}\right)$ The general term is $\tan^{-1}\left(\frac{1}{1+k(k+1)}\right)$ Using the identity $\tan^{-1}x - \tan^{-1}y = \tan^{-1}\left(\frac{x-y}{1+xy}\right)$ take $x=k+1$ and $y=k$. Then, $\tan^{-1}(k+1) - \tan^{-1}k = \tan^{-1}\left(\frac{(k+1)-k}{1+k(k+1)}\right) = \tan^{-1}\left(\frac{1}{1+k(k+1)}\right)$ So each term can be written as $\tan^{-1}\left(\frac{1}{1+k(k+1)}\right) = \tan^{-1}(k+1) - \tan^{-1}k$ Now the sum becomes $[\tan^{-1}2 - \tan^{-1}1] + [\tan^{-1}3 - \tan^{-1}2] + \dots + [\tan^{-1}(n+1) - \tan^{-1}n]$ All middle terms cancel, leaving $\tan^{-1}(n+1) - \tan^{-1}1$

Benchmark Problem 2: KCET 2026 (Q#2)

If $\sin^{-1}x + \sin^{-1}y = \left(\frac{\pi}{2}\right)$, then x^2 is equal to

- (A) $1-y^2$ [Correct]
- (B) $\sqrt{1-y^2}$
- (C) $-\sqrt{1-y^2}$
- (D) $1+y^2$

Correct Answer: A

Solution Analysis: Let $\sin^{-1}x = A$ and $\sin^{-1}y = B$ So, $A+B = \left(\frac{\pi}{2}\right)$ This gives $A = \left(\frac{\pi}{2}\right) - B$ Now, $x = \sin A$ So, $x = \sin\left(\left(\frac{\pi}{2}\right) - B\right)$ Using the identity $\sin\left(\left(\frac{\pi}{2}\right) - \theta\right) = \cos\theta$ we get $x = \cos B$ But $\sin B = y$ So, $\cos^2 B = 1 - \sin^2 B$ Hence, $x^2 = \cos^2 B = 1 - y^2$ So the correct option is $1-y^2$ Option A.

Benchmark Problem 3: KCET 2024 (Q#3)

If $2\sin^{-1}x - 3\cos^{-1}x = 4$, $x \in [-1, 1]$, then $2\sin^{-1}x + 3\cos^{-1}x$ is equal to

- (A) $(4-6\pi/5)$
- (B) $(6\pi-4/5)$
- (C) $(3\pi/2)$
- (D) 0

Correct Answer: None

Solution Analysis: $2\sin^{-1}x - 3\cos^{-1}x = 4 \Rightarrow 2\sin^{-1}x - 3\left(\frac{\pi}{2} - \sin^{-1}x\right) = 4$ [$\cos^{-1}x = \frac{\pi}{2} - \sin^{-1}x$] $\Rightarrow 5\sin^{-1}x = 4 + \frac{3\pi}{2}$ $\therefore \sin^{-1}x = \frac{4 + \frac{3\pi}{2}}{5} = \frac{8 + 3\pi}{10}$... (i) So, $2\sin^{-1}x + 3\cos^{-1}x$
$$\begin{aligned} &= 2\sin^{-1}x + 3\left(\frac{\pi}{2} - \sin^{-1}x\right) \\ &= -\sin^{-1}x + \frac{3\pi}{2} \\ &= -\frac{8 + 3\pi}{10} + \frac{3\pi}{2} = \frac{-8 - 3\pi + 15\pi}{10} \\ &= \frac{12\pi - 8}{10} = \frac{6\pi - 4}{5} \end{aligned}$$